Incremental Graph Layout by Stochastic Search: a Routed Objective, a Calibrated Metric, and a Matched-Budget Control

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Abstract

Placing the nodes of a graph in the plane with few edge crossings is NP-complete, so every layout tool runs heuristics, and the heuristics are rarely explicit about three things: what objective they optimize, whether that objective measures the drawing the reader actually sees, and whether the search spends its budget any better than uniform random sampling would. We treat all three and report what the treatment measured on eight consecutive migrations of a production database schema, each with a maintainer's accepted layout as the human reference, under a pre-registered analysis. Every budgeted search beats a centroid placement heuristic on all eight migrations (exact Wilcoxon, Holm-corrected $p = 0.023$), but uniform random sampling beats it on all eight too, so such comparisons measure a budget of evaluations against none; against the control at matched budget the genetic algorithm did not separate, the pre-registered refutation clause firing against the author's own prediction, and hill climbing is the budget-efficient method, separating from the control on seven of eight migrations, six of them by 2000 evaluations. On single-table migrations every method's margin over sampling is under three tenths of a percent, so the first deployable rule is to count the new nodes before reaching for a search. Under this objective the searches outscore the human reference, which measures the objective's blindness to what maintainers optimize rather than any superiority of the tool, and we argue this gap is why automatic graph layout has stayed bad in practice. The machinery behind the verdicts: the problem formulated incrementally (a graph grows by k nodes, the existing drawing frozen, which keeps the reader's mental map, bounds the search at $2k$ variables, and contains the classic full redraw as the limit of an anchor term); an objective that reads the drawing's real medium, orthogonally routed connectors rather than the straight center-to-center segments most objectives score, calibrated against a reference layout of certified on-screen quality with the router's shortfall printed beside every number that depends on it; and four families of stochastic search (hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform random sampling as the matched-budget control) compared by exact sign tests on paired seed panels, deterministically reproducible to the byte across platforms. One exploratory result, run after the pre-registered ones and reported as descriptive, is that the comparison between search families may be less consequential than the shape of a single proposal: letting a proposal move one node rather than all of them leaves every pre-registered run reproducing value for value, improves hill climbing on all five benchmark instances that can differ, and moves the genetic algorithm's refutation verdict by flipping exactly the instance the pre-registered analysis had identified as carrying it.

1 Introduction

Placing the nodes of a graph in the plane with the fewest edge crossings is NP-complete [1], so every tool runs heuristics. Our claim is that the practical failure is not the hardness but

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three things the heuristics leave unstated. First, the objective: what quantity, exactly, is being improved. Second, the medium: whether that quantity is measured on the drawing the reader sees. A tool that optimizes straight center-to-center segments while rendering orthogonal connectors improves a drawing nobody is shown. Third, and the one no study in graph layout runs: the control. Label placement, the adjacent field this work descends from, included a random baseline in its landmark evaluation [7], and matched-budget random controls are standard doctrine in hyperparameter search [21]; in graph layout we have not found a study that runs the control at matched budget, and without it the claim "our method improves layout" is unfalsifiable at any fixed cost.

This paper treats all three and leads with what the treatment measured. The benchmark (Section 4) is eight consecutive migrations of a production database schema, each with the maintainer's accepted layout as the human reference, scored under an analysis whose primary comparison, declared families, refutation criteria and the author's directional prediction were committed, timestamped, before any non-pilot instance was scored. The findings:

1. Every budgeted search beats the centroid placement heuristic (each new table at its neighbours' centroid) on all eight migrations, exact Wilcoxon, Holm-corrected $p = 0.023$ — and uniform random sampling beats it on all eight as well, so a method-versus-heuristic table measures a budget of evaluations against none. The comparison with teeth is against the matched-budget control.

2. Against that control the genetic algorithm did not separate ($p = 0.109$): the pre-registered refutation clause fired, against the author's own recorded prediction. Hill climbing is the budget-efficient method, directionally ahead of the genetic algorithm on every non-tied migration and separating from the control on seven of eight, six of them by 2000 evaluations.

3. The margins collapse with the task: on single-table migrations no method clears the control by more than three tenths of a percent, so the deployable rule is to count the new nodes before reaching for a search.

4. Optimizing the wrong medium has a measured price (a layout optimized under the straight-edge model retains 830 units of routed penetration on the pilot instance), and under the routed objective the searches outscore the certified human layouts, which measures the objective's blindness to what maintainers optimize rather than any superiority of the tool. We argue this gap, between what layout objectives score and what maintainers accept, is why automatic layout has stayed bad in practice.

One exploratory result is reported after those four, kept separate because it was not pre-registered and was run once they were known (Section 4.6). All four methods above move every variable on every proposal; the label-placement system this line descends from moved one label at a time. Giving the harness that choice as a parameter leaves the pre-registered runs reproducing value for value, improves hill climbing on all five benchmark instances that can differ at all, and moves the genetic algorithm's refutation verdict by flipping precisely the one instance the pre-registered analysis had already named as its hinge. The comparison this paper set out to make was between search families; the shape of their proposals appears to be the larger lever, and saying so is a claim about where to look next, not a demonstration.

The machinery that produced the verdicts comes first, because it is not specific to layout: Section 2 the four searches and the control, Section 3 the statistics. Section 4 then gives the application in full: the industry task, its name in algorithmics, the problem, the routed and calibrated objective, the benchmark of real migrations, the pre-registered experiment, and what the comparison against the human reference does and does not mean. Related work and a conclusion close.

2 The searches and the control

The harness, cjitter,[1] runs four methods behind one fitness interface: hill climbing with restarts, simulated annealing, a genetic algorithm with tournament selection and blend crossover, and uniform random sampling as the matched-budget control. Every evaluation passes through one counter, so no method can spend more than another; the budget is exact by construction, and an internal guard turns any violation into an immediate failure rather than a silently unfair comparison. All four run on the same seed panel, and the verdict against the control is an exact one-sided sign test on the paired per-seed differences: “better” requires the wins to be explainable by a fair coin with probability at most five percent, and anything else prints as “not shown,” a failure to demonstrate, not evidence of equality.

Two design choices carry the paper. First, determinism: every trajectory uses integer arithmetic and the two libm calls IEEE 754 computes exactly or rounds correctly (fabs, sqrt); the gaussian step is a sum of uniforms rather than Box–Muller, the annealing schedule uses an arithmetic exponential, and a run reproduces byte for byte across compilers and platforms, which is what lets this paper pin every reported number as a regression test. Second, the control: the first genetic algorithm shipped in this harness mutated at a fixed scale, re-scattering whatever the population had converged to, and the control's verdict read “no better than uniform sampling at equal cost.” An outside review traced the verdict to the fixed scale; with mutation decaying over the run, the same algorithm became the best method on the harness's label-placement example (90 rectangles in a container, minimum overlap). A method-versus-method table without a random control cannot distinguish a weak method from a mis-tuned one.

The three non-control methods share one more thing, which the pre-registered comparison holds fixed and Section 4.6 then varies: every proposal they make perturbs all n variables. The harness exposes that as a tuning parameter, the block, defaulting to the whole vector; everything reported before Section 4.6 is at that default.

3 The statistics

Within an instance the verdict of Section 2 is the exact one-sided sign test on the paired per-seed differences. Across instances, the pre-registered comparisons use the exact Wilcoxon signed-rank test with Holm correction over the declared family, exact McNemar on the solve-count binary, Hodges–Lehmann shifts with distribution-free confidence intervals, and a minimum detectable effect wherever a null is reported, so a null bounds rather than asserts.

One derived quantity summarizes efficiency. The *separation budget* B^* is the smallest evaluation budget at which a method beats the control on every seed of the panel, the one-sided sign test's only route to $p \leq 0.05$ at five seeds. B^* is descriptive: it detects sweeps (probability q^5 at per-seed win rate q), is censored to the tested budget grid, and a finite value can rest on tie-breaker margins, so ∞ marks non-detection at that power, never equivalence.

4 The application: a production schema and its migrations

For most of two decades the author maintained the entity–relationship diagram of a production database by hand. Every reverse-engineering of the schema replaced the diagram's positions with the tool's own placement, and restoring a 44-table diagram to readability cost about an hour each time. This is not a defect of one tool. MySQL Workbench's placement algorithm is undocumented (its manual names only the menu item), and its failure mode is on the record: bug #42600 reports a 353-table schema autoplaced with some ninety tables stacked into one pile.

[1] C99, no dependencies beyond libm; the repository carries the suites, the data and every number in this paper.

Commercial tools exist that draw entity–relationship diagrams with straight diagonal edges, a representation no maintainer draws by hand.

In algorithmics the general task is graph drawing by crossing minimization, NP-complete [1]; the version with a drawing already on screen is layout adjustment under the mental map [3]; and the version measured here is its geometric incremental instance, defined next.

4.1 The problem

Let G be a graph with a drawing D that assigns each node an axis-aligned rectangle in a bounded canvas, and let G' be a supergraph adding k nodes and their edges. The *incremental placement problem* is to position the k new rectangles, with D frozen, minimizing an objective over the resulting drawing, subject to two hard constraints enforced by construction: no two rectangles overlap, and every rectangle lies on the canvas. The search space is $2k$ real variables however large G is.

The frozen context is not a concession to tractability. The dynamic-graph-drawing literature names the reason the *mental map* [3]: a reader who has learned a drawing owns its geography, and a layout that moves what the reader knows destroys more value than it adds. The empirical evidence on mental-map preservation is mixed, helping some reading tasks and not others [18]; the migration economy is the sharper argument here, since the maintainer's acceptance is the task. Each migration adds a few tables, the diagram's owner accepts a placement, and the accepted drawing becomes the frozen context of the next migration. A version-controlled schema therefore accumulates ground truth: every revision pair is an instance, and the human's accepted answer for it.

One parameter connects this problem to the classic one. Add to the objective an anchor term $\lambda \sum_i \|p_i - p_i^0\|^2$ over the frozen nodes' displacements from their accepted positions and let them move: $\lambda \to \infty$ recovers the incremental problem, $\lambda = 0$ the full redraw, and intermediate values price the reader's geography in the objective's own units. This paper measures the $\lambda \to \infty$ end, where the practical need is; the continuum is the natural sequel, with thirty years of force-directed practice [20] as the control at the unfrozen end.

4.2 The routed objective

Entity–relationship tools draw orthogonal connectors: polylines at 0 and 90 degrees, leaving a table's border, bending once or twice. An objective over straight center-to-center segments scores penetrations and crossings that do not exist on screen and misses ones that do. We therefore route before we measure. Each edge is routed as the cheapest of 24 orthogonal candidates (two L shapes and Z shapes whose middle segment slides across the channel between the endpoints and steps beyond it), anchored on the facing borders at an attachment point unique to that edge, so no two connectors ever share a segment. Routing is sequential and aware: an edge's candidates pay for crossings against every connector already placed, so each pair of edges is priced exactly once and a route can dodge a connector instead of crossing it. The choice is deterministic, ties resolved by declaration order.

The objective is then $S = 100P + 100C + L$ over the routed drawing: penetration P as the total length of connector lying inside tables it does not join (a length, never a count, because a count is flat under small moves and gives a search nothing to follow), crossings C as a count at the same tier, and routed length L two orders of magnitude below. Node overlap and canvas containment are not terms: they are enforced by a repair callback before any candidate is scored, so no infeasible drawing can be returned as anyone's best. Optimizing the wrong medium has a measurable price: on the pilot pair, a layout optimized under the straight model retains 830 units of routed penetration where the routed optimizer reaches zero.

Orthogonal connector routing is solved to optimality in deployed libraries [10]; ours is deliberately cheap, two bends and a fixed candidate set, because it runs inside the objective at every evaluation, and the price of that choice is measured rather than assumed. A routed metric is an approximation of the tool's rendering, so we calibrate it against the one certified fact available: the maintainer's accepted layout achieved zero crossings and zero edges under tables on screen. Our router, applied to that layout, measures 27 crossings and 323 units of penetration; the shortfall, largely the gap between a two-bend router and an optimal one, is printed beside every number that depends on it, and no comparison against the human reference is read past it. The calibration also disciplined development: each router improvement (border anchors, attachment slots, crossing-aware sequencing) was accepted because this number fell, from 36 crossings and 491 penetration at the first orthogonal router to the present values. The same drawing under the straight-segment model charges the certified layout 20 crossings and 1944 penetration, which quantifies how far the diagonal representation misdescribes what the maintainer made.

4.3 The metrics

The score is the S of Section 4.2, computed identically for the straight model by substituting the direct segment for the route. Beside S:

- the *feasibility pair* (C, P), ordered lexicographically: the part of the score a reader can verify by looking at the drawing;
- the *calibration* $\chi = (C_{\mathrm{ref}}, P_{\mathrm{ref}})$, the metric applied to a reference layout whose on-screen quality is certified as zero and zero; the gap between χ and $(0, 0)$ is the metric's own error and caps every reading against the reference;
- the *transfer gap* $\Delta = S_{\mathrm{routed}}(\hat{v}_{\mathrm{straight}}) - S_{\mathrm{routed}}(\hat{v}_{\mathrm{routed}})$, the routed cost of having optimized the straight model (the pilot's penetration version appears in Section 4.2);

Realized: $n = 8$ pairs (Section 4.4), five seeds per method per pair.

4.4 The benchmark

The instances are real. One production schema's version history holds seventeen revisions of its MySQL Workbench model, sixteen consecutive pairs, each a migration whose diagram a maintainer restored by hand and accepted on both sides. Per pair, the task is the previous revision's diagram frozen as drawn, the added tables to place, and the maintainer's accepted coordinates in the next revision as the human reference; the median displacement of surviving tables between revisions, 36 to 1254 canvas units across the usable pairs, is recorded per pair, since the human answered in a context that sometimes moved. The pre-registered low-displacement subset (at most 20 units) is empty, so that confound on the human reference is uncontrolled here. Seven of the sixteen pairs added no tables (documentation edits, pure re-layouts) and are excluded with that reason; one pair served as the pilot and is excluded from the primary analysis; eight remain. Two threats are recorded rather than repaired: matching is by table name, so a migration that renames tables can count renames as additions (pair 14, "refactoring and renaming," is the flagged case); and the eight pairs are consecutive migrations of one schema, each frozen context containing the previous pair's answer, where the exact tests assume independent pairs, so generalization beyond this database is external validity, not statistics.

The current pair ships with the harness as its example, anonymized for publication: every table renamed to a letter, business column names neutralized down into the view SQL, the machine identifier that v1-style UUIDs embed scrubbed, and the geometry, which is the ground truth, untouched. The mapping back to the schema is deliberately stored nowhere.

Figure 1: One benchmark instance, anonymized. Left: the frozen diagram before the migration (36 tables). Right: after it, the ten added tables in amber, connectors routed orthogonally. The maintainer's accepted coordinates for the added tables are the human reference.

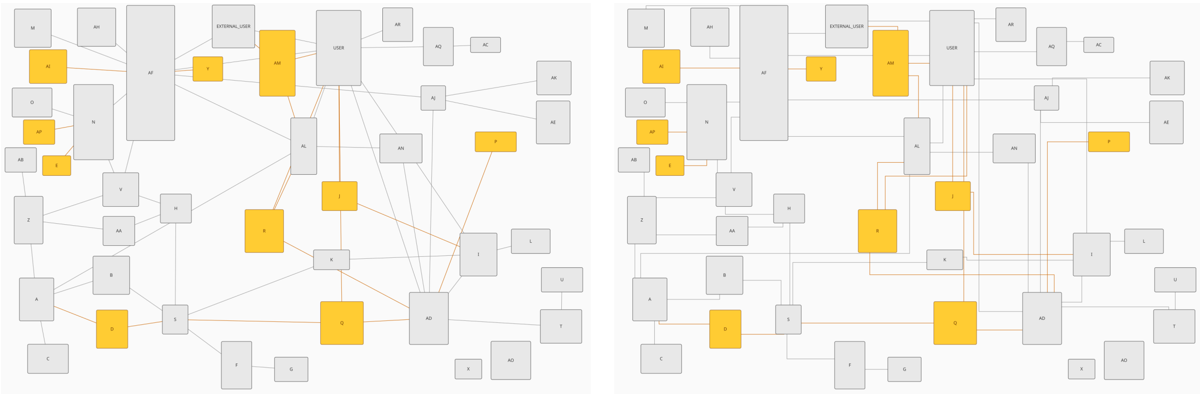

Figure 2: The same drawing under the two edge models: straight center-to-center segments (left) and routed orthogonal connectors (right), what the maintainer's tool draws and this paper's objective reads.

4.5 The pre-registered experiment

The analysis was committed, timestamped, before any non-pilot pair was scored: primary comparison, declared families, refutation criterion, and the author's own directional prediction. Per pair and method, five seeds at the shipped budget of 8000 evaluations (with 500, 2000 and 32000 for the separation-budget table), the per-pair value the median over seeds; the statistics of Sections 3 and 4.3 throughout.

The primary comparison succeeded, and we state at once how far it reaches. The genetic algorithm beats the centroid heuristic (each new table at its neighbours' centroid) on eight of eight migrations, exact Wilcoxon $p = 0.0078$, Holm-corrected 0.023, Hodges–Lehmann shift $-72{,}287$ $[-150{,}515, -20{,}728]$; hill climbing reads the same; and so does uniform random sampling, which also beats the centroid on all eight, so these comparisons measure a budget of evaluations against none. At $n = 8$ the exact test's 0.0078 is its smallest attainable value, an eight-for-eight sweep, so the Holm-corrected 0.023 carries no margin. On the pre-registered feasibility order (C, P) the direction reverses: the centroid beats the GA on four pairs against one (three ties) and hill climbing three against two, the searches buying zero penetration at the price of crossings; neither direction approaches significance, and Figure 3's "loses everywhere" is a statement about S alone. The Friedman omnibus over the four methods rejects at $p \approx 0.0005$ with mean ranks climb 1.44, anneal 2.19, GA 2.69, random 3.69; the Nemenyi post-hoc promised in the pre-registration is omitted as redundant to the reported pairwise family, a declared deviation.

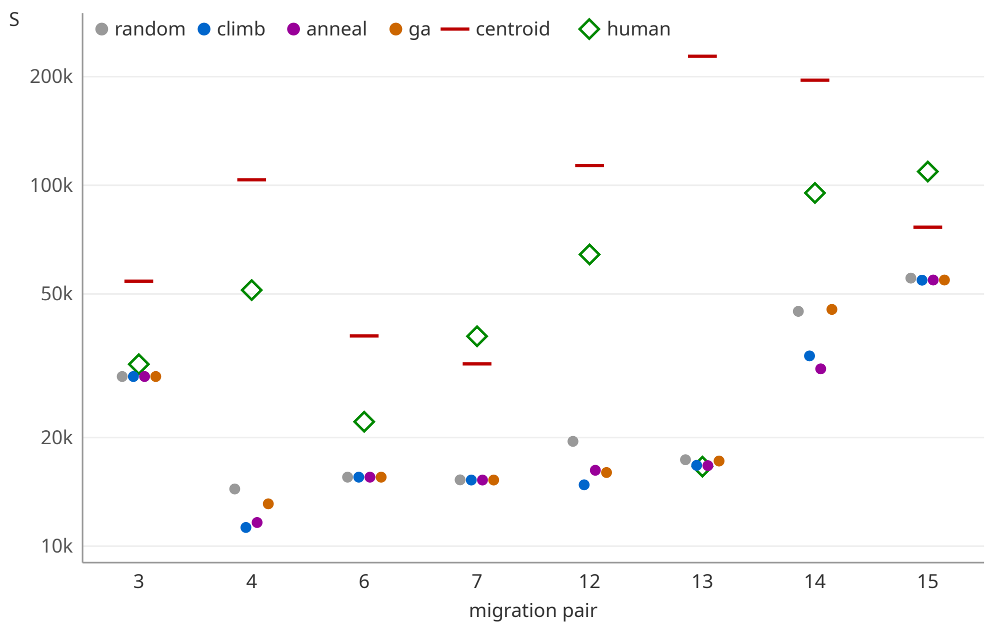

Figure 3: Per-pair medians of the routed objective at budget 8000 (five seeds), log scale, with the centroid heuristic and the human reference. On the single-table pairs the four methods nearly coincide; the heuristic loses everywhere on S (on the feasibility order the comparison reverses; Section 4.5); the human reference carries the router's calibration error and is a reference, never a target.

Table 1: Separation budget B^* per pair: the smallest budget (of 500, 2000, 8000, 32000) at which the five-seed sign test against the matched-budget control reaches $p \leq 0.05$; ∞ means not detected at any tested budget. The median column ranks ∞ above every finite budget; the never column counts the ∞ entries.

pair	3	4	6	7	12	13	14	15	median	never
climb	∞	2000	2000	2000	2000	8000	2000	500	2000	1
anneal	∞	2000	∞	32000	500	8000	500	500	5000	2
ga	∞	32000	2000	500	500	∞	∞	8000	20000	3

Two declared clauses then fired against the experimenters, which we report with the same prominence. The refutation clause: the GA did not separate from uniform random sampling. The decomposition matters: six wins, one tie (the fully degenerate pair 3), and one outright loss to the control on pair 14, the rename-flagged pair, for exact Wilcoxon $p = 0.109$; flipping that single pair would give $p = 0.016$, so the verdict is carried by one loss on a suspect instance and by n. The easy pairs are real but they are not the whole story: on the three single-table migrations every method's margin over the control is at most three tenths of a percent (pair 3 is identical to five digits), while the one $k = 2$ pair shows a genuine 3.5 percent search win, so the deployable rule is stated for $k = 1$ and by effect size: count the new nodes before reaching for a search.

And the author's pre-registered prediction, that the GA would be the most efficient search, was refuted by its own experiment. Hill climbing is directionally ahead of the GA on every non-tied migration (seven of seven, exact $p = 0.016$, Holm 0.031); the tally's robustness has a limit we state rather than hide: two of those wins are margins of 0.02 and 1.6 score units on the near-degenerate pairs, below the within-pair seed spread of about 27, and treating them as ties gives five of five, $p = 0.0625$, so the ordering is directionally consistent and not robust to a tie threshold the pre-registration failed to declare. The declared family's other half is reported with

Table 2: Per-pair medians of the routed objective at budget 8000, whole-vector proposals $\rightarrow$ one table per proposal, on the five pre-registered pairs with $k \geq 2$. The three $k = 1$ pairs are identical by construction and omitted. The control is unaffected by the block, as it must be: uniform sampling makes no proposals.

pair	k	climb		anneal		ga		random
		$2k$	2	$2k$	2	$2k$	2	
4	5	11266	10459	11623	10485	13091	12459	14394
12	5	14778	14776	16224	15398	15986	15830	19500
13	2	16745	16656	16707	16700	17210	17218	17347
14	4	33653	30784	30985	31209	45277	34208	44749
15	4	54560	54494	54633	54584	54642	54632	55308

it: GA versus annealing is not shown ($p = 0.297$). Climbing separates from the control on seven of eight pairs, six by 2000 evaluations (the GA: five of eight, three by 2000; medians in Table 1); an exploratory bound finds climbing and annealing indistinguishable to within a tenth of the primary effect. The pilot pair, with $k = 10$, had pointed the other way, GA first; the surviving hypothesis, that recombination pays only when there are enough modules to recombine, is the successor pre-registration, and it will be scored the same way this one was.

4.6 Exploratory: the shape of the proposal

Everything above compares search *families* at one fixed proposal shape: each of the four methods, on every step, perturbs all $2k$ variables at once. That was never argued for, and it is not what the label-placement system this line descends from did in 2001, which moved one label at a time under the one summed cost. We therefore added a single tuning parameter, the *block*: the number of variables one proposal may move, taken in blocks that tile the vector and cycle, with the whole vector as the default. Setting it to 2 makes each proposal move one table.

The analysis in this subsection is exploratory. It was not pre-registered, it was run after the pre-registered results above were known, and its p-values are descriptive. What it does not do is disturb them: at block $= 2k$ all 640 runs of the pre-registered sweep reproduce value for value, so the parameter is an addition to the harness rather than a revision of it.

Two structural facts bound what the benchmark can say. Three of the eight pairs add a single table, where $2k = 2$ and a block of 2 *is* the whole vector, so those instances are identical by construction; only five pairs can differ at all. At $n = 5$ the exact two-sided Wilcoxon test's smallest attainable value is 0.0625, so no result here can reach 0.05 two-sided, whatever the data.

On those five pairs, blocked proposals improve hill climbing on five of five (Table 2), exact two-sided $p = 0.0625$ — a clean sweep sitting exactly on the power floor — with a Hodges–Lehmann shift of -436 score units. Annealing improves on four of five ($p = 0.31$) and the genetic algorithm on four of five ($p = 0.125$). The separation budgets of Table 1 do not clearly improve: climbing's median B^* is 2000 under both shapes, three pairs falling and one rising.

The one verdict that moves is the refutation clause. Against the matched-budget control at budget 8000 the genetic algorithm goes from $p = 0.109$ to $p = 0.0156$, and it does so by flipping exactly the instance Section 4.5 and the appendix both identified as the hinge: pair 14, the rename-flagged pair, where the GA loses to the control at whole vector proposals (45,277 against 44,749) and beats it at block 2 (34,208). We report this as what it is. The pre-registered verdict stands as recorded; the exploratory finding is that it was a verdict about a method family *at one proposal shape*, and that the shape was the larger lever.

The margins in Table 2 are mostly small, between 0.01 and 8.5 percent, and they do not order by k within this range. Two instances outside the pre-registered eight show what the parameter does where there is more to decompose, and we quote them as illustrations rather than as evidence. On the harness's packing example, 90 rectangles at 36 percent area coverage, whole-vector proposals leave climbing at a median overlap of 12.8, annealing at 128 and the GA at 1.38 after 20,000 evaluations; at block 2 all three reach exactly zero on all seven seeds, the clean layout, which no method reaches at any budget we tried with whole-vector proposals. On the $k = 10$ instance shipped with the harness, climbing's median falls from 46,134 to 32,293 and its range across five seeds falls from 18,662 to 178: the blocked search returns substantially the same drawing from every seed, which for a tool run once matters more than the median.

The mechanism we propose, and do not claim to have established, is that incremental placement is a sum over objects that interact weakly — in the shipped instance no edge joins two added tables at all, so the k placements couple only through overlap and crossings. On such an objective a whole-vector proposal that seats one object well and another badly is rejected for the second, and the acceptance rate collapses: on the shipped instance climbing accepts 47 of 8000 proposals at whole vector and 124 at block 2. If that is right, the block's value grows with the number of weakly coupled objects, which would put the benchmark's $k \leq 5$ instances near the bottom of its range and is the same k-dependence the pre-registered analysis found for search-versus-control margins. Testing it needs instances with larger k than this schema's migrations supply, and it is declared as such rather than concluded.

4.7 The human comparison

Under the routed objective the searches outscore the maintainer's accepted layouts on most pairs. Read through the calibration line, this is not a claim that the tool out-draws a person. Much of the human reference's score is the router failing to reproduce connectors the maintainer's tool routed cleanly; and the feasibility pairs disagree with the scores, the searches reaching zero penetration while the human's on-screen crossings stay lower. Person and tool jointly achieved zero and zero; each automated party wins a different half of that, and the router owns the remaining gap.

A layout objective of this family sees crossings, penetration and length; a maintainer optimizes semantic grouping, aligned rows and room to grow. Any such objective will therefore beat the human on itself and lose on the screen, and a tool guided only by such an objective ships the drawing the reader then spends an hour repairing. The gap between drawing metrics and human judgment has an empirical literature of its own [17]; what the migration benchmark adds is the maintainer's acceptance as the recorded outcome. That gap, between what layout objectives score and what maintainers accept, is our explanation for why automatic diagram layout has stayed bad in practice despite methods that demonstrably beat their controls: the searches are fine; the objectives do not yet describe the job. Objectives anchored to certified human layouts, with their approximation error measured and printed, are the repair this paper proposes.

5 Related work

Crossing minimization is NP-complete [1]; orthogonal drawing with bend minimization is a literature of its own [4], founded industrially on exactly our application, the orthogonal drawing of database diagrams [5]. Stochastic search for layout begins in earnest with Davidson and Harel's simulated annealing over aesthetic criteria [2] and continues through genetic approaches such as TimGA [6]; label placement ran the landmark empirical comparison, random baseline included [7]. The mental map was named by Misue, Eades, Lai and Sugiyama [3], dynamic graph drawing is surveyed in [19], and geometric incremental placement has direct precedents: interactive orthogonal drawing with no-change insertion scenarios [8], and Dunnart's constrained interactive layout with routed connectors and a preserved drawing [9]. Orthogonal connector routing is solved to optimality in deployed libraries [10]. On the layered side, the Constrained Incremental Graph Drawing Problem carries the metaheuristics line [11, 12], most recently hybridized with graph representation learning [13], over synthetic instance sets. Force-directed practice is surveyed in [20]; graphviz's `neato`, honoring pinned positions, is the baseline the anchor-term sequel must face. The empirical gap between drawing metrics and human judgment is Purchase's line [17].

Against that background the novelty claim is narrow and we state it narrowly. Interactive tools place into frozen drawings and route connectors [8, 9]; evaluations have included random baselines [7]; each ingredient exists somewhere. Outside layout, head-to-head comparisons of exactly these search methods are abundant ([14] is an early one; [15] found the simple methods winning), and a uniform-random baseline at matched budget is prescribed practice in search-based software engineering [16], so the harness's control is an application of doctrine, not an invention. What we have not found together is the evaluation half: an objective calibrated against a certified human layout with its error printed, a random control at matched budget in graph layout, exact paired inference over real migration instances with human answers, and a pre-registration whose refutation clauses were allowed to fire, one of them against the author.

6 Conclusion

An hour of hand repair per reverse-engineering, for years, was a measurement nobody wrote down. Written down properly, it became: a formulation (frozen context, $2k$ variables, an anchor term to the classic problem), an objective that reads routed connectors with its calibration error measured and printed, a harness in which four searches spend identical budgets against a random control and are judged by exact paired tests, a benchmark of eight real migrations with human-accepted answers, and a pre-registered experiment that ranked the searches above the centroid heuristic, refuted the author's own prediction, and produced one deployable rule, from eight migrations of one schema and stated with that scope: count the new nodes first. One of the successor questions has since been answered inside the harness rather than left waiting: the per-label climber that started this line in 2001 does hold its own, and more than that (Section 4.6). Expressed as a block size on the proposal it improves hill climbing on every benchmark instance that can differ, solves the packing example outright where whole-vector proposals never do, and moves the one verdict this paper reported against its author. We draw the methodological moral rather than the layout one: a comparison between search families holds the proposal shape fixed without saying so, and a parameter nobody varied turned out to matter more than the parameter everybody did. The questions still declared and waiting are whether recombination's advantage returns with the module count, whether the block's value grows with the number of weakly coupled objects as we suspect and this benchmark is too small to show, and at what objective noise any search stops being worth its budget.

Appendix: leave-one-out robustness

The eight pairs are consecutive migrations of one schema, so the exact tests' independence assumption is stated rather than satisfied (Section 4.4). As a post-hoc sensitivity check (the pre-registered analysis is Section 4.5), Table 3 recomputes each declared comparison's exact two-sided Wilcoxon signed-rank p, on the per-pair medians at budget 8000 with zero differences dropped, once on all eight pairs and once with each pair excluded. Three readings. Every search-versus-centroid sweep survives every exclusion (a seven-pair sweep's exact p is 0.016). Hill climbing's two verdicts, against the genetic algorithm and against the control, stay at or below 0.031 under every exclusion. The genetic algorithm against the control moves between 0.031 and 0.219: excluding the rename-flagged pair 14 is the one exclusion under which the GA would separate, which confirms, by exclusion rather than by the sign flip reported in Section 4.5, that the refutation verdict is carried by that single suspect instance and by n.

Table 3: Exact two-sided Wilcoxon signed-rank p per comparison: all eight pairs, then with the named pair excluded.

comparison	all	−3	−4	−6	−7	−12	−13	−14	−15
GA vs. centroid	0.0078	0.016	0.016	0.016	0.016	0.016	0.016	0.016	0.016
climb vs. centroid	0.0078	0.016	0.016	0.016	0.016	0.016	0.016	0.016	0.016
random vs. centroid	0.0078	0.016	0.016	0.016	0.016	0.016	0.016	0.016	0.016
GA vs. random	0.1094	0.109	0.219	0.156	0.156	0.219	0.156	0.031	0.219
climb vs. GA	0.0156	0.016	0.031	0.031	0.031	0.031	0.031	0.031	0.031
climb vs. random	0.0156	0.016	0.031	0.031	0.031	0.031	0.031	0.031	0.031

Reproducibility

Every number in this paper regenerates from the repository at the frozen commit: the per-seed data, the extraction and analysis code, the pre-registration with its addenda and this paper's source live together, and the harness's own test suite pins the shipped example's every printed digit. The exploratory arm of Section 4.6 was run on a later harness carrying the block parameter, and the reproduction is the check that licenses comparing the two: re-running the whole pre-registered sweep on that harness with the block at its default reproduces all 640 per-seed values of the frozen `results.csv`, so the two arms differ in the parameter and in nothing else. The benchmark geometry derives from a private schema and is carried anonymized; the anonymization mapping is stored nowhere.

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